

Brutal Practice Paper B08 - Motion & Time / Large Numbers / Wastewater / Decimals

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. A car covers 180 km in 2.5 hours at a steady rate. What is its speed in metres per second? [\[Motion & Time\]](#)
 - (A) 200 m/s
 - (B) 25.2 m/s
 - (C) 15 m/s
 - (D) 20 m/s
 - (E) 16 m/s
 - (F) 72 m/s

2. Write the number "fifty lakh sixty thousand seven hundred eight" in figures. [\[Large Numbers\]](#)
 - (A) 5,06,708
 - (B) 5,60,708
 - (C) 50,60,708
 - (D) 50,60,078
 - (E) 50,06,708
 - (F) 50,60,780
 - (G) 5,00,60,708

3. Sewage is about 99.9% water and the rest is contaminants. In 1000 litres of sewage, how many litres are contaminants? [\[Wastewater Story\]](#)
 - (A) 11 litres
 - (B) 1 litre
 - (C) 0.1 litre
 - (D) 99.9 litres
 - (E) 100 litres
 - (F) 10 litres
 - (G) 999 litres

4. Find the value of $23.6 + 4.85 + 0.073$. [\[A Peek Beyond the Point\]](#)
 - (A) 28.52
 - (B) 285.23
 - (C) 28.623
 - (D) 28.533
 - (E) 28.523
 - (F) 28.43
 - (G) 27.523

5. A train moves at a steady 90 km/h. How many metres does it travel in 12 seconds? [\[Motion & Time\]](#)
- (A) 1080 m
 - (B) 250 m
 - (C) 30 m
 - (D) 300 m
 - (E) 360 m
 - (F) 108 m
 - (G) 270 m
6. In the number 7,84,06,295, what is the place value of the digit 4? [\[Large Numbers\]](#)
- (A) 4,000,000
 - (B) 4,00,000
 - (C) 4,000
 - (D) 40,00,000
 - (E) 4,00,00,000
 - (F) 4
 - (G) 40,000
7. Which is the correct order of the main steps in a wastewater treatment plant? [\[Wastewater Story\]](#)
- (A) Aeration first, then bar screens, then grit removal
 - (B) Bar screens remove large objects, grit and sand settle out, a clarifier removes sludge, then air and bacteria clean the water
 - (C) Biogas is collected first, then large objects screened
 - (D) Sand settling first, then bar screens, then clarifier
 - (E) Sludge is removed first, then large objects are screened out last
 - (F) Chlorine is added first, then bar screens, then sand settles
 - (G) The water is chlorinated, then released, then screened
8. What is the value of 0.6×0.05 ? [\[A Peek Beyond the Point\]](#)
- (A) 0.003
 - (B) 0.30
 - (C) 0.0003
 - (D) 0.3
 - (E) 0.06
 - (F) 3
 - (G) 0.03
9. A simple pendulum makes 30 complete oscillations in 45 seconds. What is its time period? [\[Motion & Time\]](#)
- (A) 1 s
 - (B) 15 s
 - (C) 0.67 s
 - (D) 2 s
 - (E) 75 s
 - (F) 1350 s
 - (G) 1.5 s

10. The number 30,500,000 reads "thirty million five hundred thousand" in the International system. How is it read in the Indian system? [\[Large Numbers\]](#)

- (A) thirty-five lakh
- (B) three crore five lakh
- (C) three lakh five crore
- (D) three crore fifty lakh
- (E) thirty lakh five thousand
- (F) thirty crore five lakh
- (G) three crore five thousand

11. A treatment plant cleans 24,00,000 litres of sewage every day. How many litres does it clean in 30 days? [\[Wastewater Story\]](#)

- (A) 7,20,000 L
- (B) 72,00,000 L
- (C) 24,00,030 L
- (D) 2,40,00,000 L
- (E) 72,00,00,000 L
- (F) 7,02,00,000 L
- (G) 7,20,00,000 L

12. What is 4.5 divided by 0.9? [\[A Peek Beyond the Point\]](#)

- (A) 4
- (B) 0.2
- (C) 5
- (D) 0.5
- (E) 50
- (F) 4.5

13. A cyclist rides 30 km to a town at 15 km/h and returns the same 30 km at 10 km/h. What is the average speed for the whole trip? [\[Motion & Time\]](#)

- (A) 24 km/h
- (B) 12.5 km/h
- (C) 60 km/h
- (D) 6 km/h
- (E) 12 km/h
- (F) 5 km/h
- (G) 25 km/h

14. Estimate $6,12,478 + 3,87,945$ by first rounding each number to the nearest ten thousand, then adding. [\[Large Numbers\]](#)

- (A) 9,00,423
- (B) 9,00,000
- (C) 10,00,000
- (D) 10,00,423
- (E) 11,00,000
- (F) 9,90,000
- (G) 10,10,000

15. A family keeps tipping leftover cooking oil and fats into the kitchen drain. Why is this a bad idea for the sewer pipes? [\[Wastewater Story\]](#)

- (A) They feed helpful bacteria and speed up treatment
- (B) They harden and stick inside the pipes, blocking the flow
- (C) They are harmless because they are food
- (D) They soften the water in the pipes
- (E) They turn the water clear
- (F) They make biogas form faster
- (G) They clean the inside of the pipes

16. Convert the fraction $\frac{3}{8}$ into a decimal. [\[A Peek Beyond the Point\]](#)

- (A) 2.67
- (B) 0.375
- (C) 0.125
- (D) 0.83
- (E) 0.38
- (F) 0.38

17. A bus's odometer reads 45670 km at 8:00 am and 45920 km at 1:00 pm. What was its average speed?

[\[Motion & Time\]](#)

- (A) 91590 km/h
- (B) 100 km/h
- (C) 41.67 km/h
- (D) 83.33 km/h
- (E) 25 km/h
- (F) 250 km/h
- (G) 50 km/h

18. A factory makes 24,750 bottles each day. How many bottles does it make in the whole month of June?

[\[Large Numbers\]](#)

- (A) 24,780
- (B) 74,25,000
- (C) 7,42,050
- (D) 7,42,500
- (E) 74,250
- (F) 7,67,250

19. The treated water leaving a plant still has harmful microbes. Which is commonly added to kill them before the water is released? [\[Wastewater Story\]](#)

- (A) cooking oil
- (B) chlorine
- (C) lime juice
- (D) sugar
- (E) table salt
- (F) extra sludge

20. Among 0.7, 0.07, 0.707 and 0.077, which is the greatest? [\[A Peek Beyond the Point\]](#)

- (A) 0.0707
- (B) 0.707
- (C) 0.07
- (D) 0.077
- (E) 0.7
- (F) 0.077
- (G) 0.770

21. A distance-time graph is a straight line passing through the points (0 s, 0 m) and (4 s, 60 m). What is the speed, and what does the straight line tell us? [\[Motion & Time\]](#)

- (A) 240 m/s, uniform speed
- (B) 4 m/s, uniform speed
- (C) 15 m/s, the object moves at uniform (constant) speed
- (D) 0.067 m/s, uniform speed
- (E) 30 m/s, changing speed
- (F) 15 m/s, the object moves with changing speed
- (G) 60 m/s, uniform speed

22. Which number is the greatest: 9,09,990 or 90,99,099 or 9,90,099 or 90,90,990? [\[Large Numbers\]](#)

- (A) 9,09,990
- (B) 90,99,099
- (C) 90,90,990
- (D) 99,00,000
- (E) 90,09,990
- (F) 9,90,099
- (G) 9,99,099

23. A leaking tap wastes 1.5 litres of clean water every minute. How many litres are wasted in one full day?

[\[Wastewater Story\]](#)

- (A) 90 L
- (B) 2160 L
- (C) 21600 L
- (D) 36 L
- (E) 216 L
- (F) 2150 L
- (G) 1440 L

24. A pen costs Rs 12.50. If you buy 8 pens and pay with a Rs 200 note, how much change do you get? [\[A](#)

[Peek Beyond the Point\]](#)

- (A) Rs 88
- (B) Rs 100
- (C) Rs 104
- (D) Rs 108
- (E) Rs 92
- (F) Rs 90

25. A train 200 m long passes a standing pole in 8 seconds. What is its speed in km/h? [\[Motion & Time\]](#)

- (A) 90 km/h
- (B) 25 km/h
- (C) 0.04 km/h
- (D) 1600 km/h
- (E) 8 km/h
- (F) 72 km/h

26. Which number equals $8 \times 10,00,000 + 5 \times 10,000 + 3 \times 100 + 7$? [\[Large Numbers\]](#)

- (A) 8,05,307
- (B) 80,53,007
- (C) 85,00,307
- (D) 80,50,307
- (E) 80,05,307
- (F) 80,50,370
- (G) 80,50,037

27. A town produces 5,00,000 litres of sewage daily. The plant recovers 80% of it as treated water for gardens. How many litres are reused each day? [\[Wastewater Story\]](#)

- (A) 4,00,000 L
- (B) 5,80,000 L
- (C) 1,00,000 L
- (D) 4,00,00,000 L
- (E) 4,50,000 L
- (F) 80,000 L
- (G) 40,000 L

28. Multiply 5.24 by 1000. [\[A Peek Beyond the Point\]](#)

- (A) 5240.0
- (B) 5240
- (C) 0.00524
- (D) 52.4
- (E) 52400
- (F) 5.240
- (G) 524

29. A pendulum takes 2.5 seconds for one complete oscillation. How many complete oscillations does it make in 1 minute? [\[Motion & Time\]](#)

- (A) 0.04
- (B) 150
- (C) 24
- (D) 2.5
- (E) 30
- (F) 25
- (G) 20

- 30.** A book has 1,250 pages. A library owns 3,840 copies of this book. How many printed pages are there in all? [\[Large Numbers\]](#)
- (A) 48,00,000
 - (B) 48,000
 - (C) 50,90,000
 - (D) 5,090
 - (E) 4,08,00,000
 - (F) 4,80,00,000
 - (G) 4,80,000
- 31.** In a settling tank, what is the name of the solid material that sinks to the bottom, and what useful gas can it later produce? [\[Wastewater Story\]](#)
- (A) sludge, which produces oxygen
 - (B) grit, which produces chlorine
 - (C) scum, which produces biogas
 - (D) clarified water, which produces steam
 - (E) sludge, which can produce biogas
 - (F) foam, which produces nitrogen
 - (G) scum, which produces oxygen
- 32.** What is 78.5 divided by 100? [\[A Peek Beyond the Point\]](#)
- (A) 785
 - (B) 0.785
 - (C) 7850
 - (D) 0.0785
 - (E) 0.00785
 - (F) 7.85
- 33.** A sprinter runs at 9 m/s. Express this speed in km/h. [\[Motion & Time\]](#)
- (A) 90 km/h
 - (B) 3.24 km/h
 - (C) 162 km/h
 - (D) 2.5 km/h
 - (E) 9 km/h
 - (F) 32.4 km/h
 - (G) 45 km/h
- 34.** By how much is the greatest 6-digit number larger than the greatest 5-digit number? [\[Large Numbers\]](#)
- (A) 9,00,000
 - (B) 1,00,000
 - (C) 10,00,000
 - (D) 90,000
 - (E) 8,99,999
 - (F) 9,99,999
 - (G) 9,90,000

35. Open drains and pools of stagnant sewage breed mosquitoes. If one drain produces 250 mosquitoes a day, how many does it produce in the 31 days of a month? [\[Wastewater Story\]](#)

- (A) 7,500
- (B) 77,500
- (C) 8,050
- (D) 281
- (E) 7,750
- (F) 750

36. A 2.5 m ribbon has two pieces cut from it, of lengths 0.45 m and 1.2 m. How much ribbon is left? [\[A Peek Beyond the Point\]](#)

- (A) 1.05 m
- (B) 0.85 m
- (C) 0.75 m
- (D) 0.65 m
- (E) 0.8 m
- (F) 4.15 m
- (G) 0.95 m

37. A car travels 100 km in the first 2 hours and 120 km in the next 3 hours. What is its average speed for the whole journey? [\[Motion & Time\]](#)

- (A) 44 km/h
- (B) 48 km/h
- (C) 22 km/h
- (D) 45 km/h
- (E) 220 km/h
- (F) 40 km/h

38. Estimate the product 312×489 by rounding each number to the nearest hundred, then multiplying. [\[Large Numbers\]](#)

- (A) 1,40,000
- (B) 1,45,000
- (C) 15,000
- (D) 1,55,000
- (E) 1,52,568
- (F) 1,50,000

39. What is a "sewer"? [\[Wastewater Story\]](#)

- (A) a tank that stores clean drinking water
- (B) a chemical that cleans water
- (C) the underground pipe network that carries sewage away
- (D) a worker who repairs taps
- (E) the dirty waste water itself
- (F) a machine that pumps air
- (G) the solid sludge at a plant

40. Find 0.25 of 84. [\[A Peek Beyond the Point\]](#)

- (A) 8.4
- (B) 210
- (C) 25
- (D) 16.8
- (E) 33.6
- (F) 21
- (G) 2.1

41. A plane flies 2450 km at a steady 700 km/h. How long does the flight take? [\[Motion & Time\]](#)

- (A) 2 h 30 min
- (B) 3 h 45 min
- (C) 3 h 5 min
- (D) 4 h 10 min
- (E) 3 h 50 min
- (F) 3 h 30 min
- (G) 35 min

42. A printing press prints 8,500 newspapers each hour and runs for 6 hours every day. How many newspapers does it print in a week (7 days)? [\[Large Numbers\]](#)

- (A) 3,07,000
- (B) 3,50,700
- (C) 3,57,000
- (D) 357
- (E) 59,500
- (F) 35,70,000
- (G) 51,000

43. Which of these should NOT be flushed down a toilet because it does not break down and chokes the system? [\[Wastewater Story\]](#)

- (A) mild liquid soap
- (B) small amounts of urine
- (C) human waste
- (D) plastic wrappers and sanitary items
- (E) a little hand wash
- (F) water

44. Express the decimal 0.625 as a fraction in its lowest terms. [\[A Peek Beyond the Point\]](#)

- (A) $\frac{5}{16}$
- (B) $\frac{625}{1000}$
- (C) $\frac{2}{3}$
- (D) $\frac{5}{8}$
- (E) $\frac{5}{80}$
- (F) $\frac{1}{16}$

45. A simple pendulum is timed for 20 oscillations three times: 30 s, 31 s and 29 s. Using the average of these readings, find the time period. [\[Motion & Time\]](#)
- (A) 90 s
 - (B) 1.45 s
 - (C) 1.55 s
 - (D) 0.67 s
 - (E) 2 s
 - (F) 1.5 s
 - (G) 1 s
46. A satellite orbits 36,00,000 metres above the Earth. Express this height in kilometres. [\[Large Numbers\]](#)
- (A) 36,000 km
 - (B) 360 km
 - (C) 3,600 km
 - (D) 3,60,00,000 km
 - (E) 36 km
 - (F) 3,600,000 km
 - (G) 3,60,000 km
47. What do the "bar screens" at the start of a treatment plant remove from the sewage? [\[Wastewater Story\]](#)
- (A) harmful bacteria
 - (B) the colour of the water
 - (C) large floating objects like rags, sticks and cans
 - (D) the bad smell
 - (E) tiny air bubbles
 - (F) fine sand and grit
48. What is 12.5 divided by 0.25? [\[A Peek Beyond the Point\]](#)
- (A) 12.5
 - (B) 3.125
 - (C) 31.25
 - (D) 0.5
 - (E) 50
 - (F) 500
 - (G) 5
49. A toy car moves at a steady 0.5 m/s. Express this speed in km/h. [\[Motion & Time\]](#)
- (A) 1.8 km/h
 - (B) 0.139 km/h
 - (C) 5 km/h
 - (D) 0.28 km/h
 - (E) 0.5 km/h
 - (F) 2.5 km/h
 - (G) 9 km/h

50. A number, when rounded to the nearest thousand, becomes 56,000. What is the largest number it could have been? [\[Large Numbers\]](#)

- (A) 56,499
- (B) 55,999
- (C) 57,000
- (D) 56,500
- (E) 56,449
- (F) 56,549
- (G) 56,999

51. After treatment, what is the cleaned water usually released into? [\[Wastewater Story\]](#)

- (A) the sludge tank
- (B) drinking-water taps directly
- (C) back into the sewers
- (D) the bar screens again
- (E) rivers, streams or the sea
- (F) sealed bottles for sale

52. Round the number 7.864 to two decimal places. [\[A Peek Beyond the Point\]](#)

- (A) 7.87
- (B) 7.86
- (C) 7.85
- (D) 7.864
- (E) 7.9
- (F) 8.86
- (G) 7.88

53. A train 120 m long, moving at 54 km/h, crosses a bridge 180 m long. How long does it take to completely cross the bridge? [\[Motion & Time\]](#)

- (A) 300 s
- (B) 16.2 s
- (C) 20 s
- (D) 5.56 s
- (E) 8 s
- (F) 12 s
- (G) 24 s

54. In the number 5,55,555, by how much does the value of the 5 in the lakhs place exceed the value of the 5 in the hundreds place? [\[Large Numbers\]](#)

- (A) 5,00,500
- (B) 4,99,050
- (C) 4,99,500
- (D) 4,99,995
- (E) 5,00,000
- (F) 4,95,000
- (G) 5,55,000

55. Dried sludge from a treatment plant, once fully decomposed, is most commonly used as what?

[\[Wastewater Story\]](#)

- (A) chlorine gas
- (B) drinking water
- (C) manure (fertiliser) for plants
- (D) fuel for cars
- (E) cooking oil
- (F) window glass
- (G) table salt

56. Round the number 3.196 to one decimal place. [\[A Peek Beyond the Point\]](#)

- (A) 3.2
- (B) 4.2
- (C) 3.3
- (D) 3.1
- (E) 3.196
- (F) 3.19
- (G) 3.0

57. A car covers 60 km in the first hour, 70 km in the second hour and 50 km in the third hour. What is its average speed? [\[Motion & Time\]](#)

- (A) 90 km/h
- (B) 60 km/h
- (C) 26.67 km/h
- (D) 50 km/h
- (E) 55 km/h
- (F) 180 km/h
- (G) 70 km/h

58. A stadium seats 65,000 people. It is completely full for 4 matches, but at a 5th match only 48,750 people attend. What is the total attendance over the 5 matches? [\[Large Numbers\]](#)

- (A) 3,08,570
- (B) 2,60,000
- (C) 3,25,000
- (D) 3,13,750
- (E) 30,87,50
- (F) 2,96,250
- (G) 3,08,750

59. A community toilet block serves 480 people. Each person uses 6 litres of water per visit and visits twice a day. How much water is used daily? [\[Wastewater Story\]](#)

- (A) 486 L
- (B) 2,880 L
- (C) 5,760 L
- (D) 5,780 L
- (E) 57,600 L
- (F) 2,882 L
- (G) 5,660 L

- 60.** A rod is 1.75 m long. A second rod is 0.8 m longer than the first. What is the total length of both rods together? [\[A Peek Beyond the Point\]](#)
- (A) 4.35 m
 - (B) 4.25 m
 - (C) 1.8 m
 - (D) 4.30 m
 - (E) 2.55 m
 - (F) 4.55 m
 - (G) 3.55 m
- 61.** An object covers 4 m in the first 2 seconds and 12 m in the next 2 seconds. What is its average speed over the 4 seconds, and is its motion uniform? [\[Motion & Time\]](#)
- (A) 4 m/s, motion is uniform
 - (B) 4 m/s, motion is non-uniform
 - (C) 3 m/s, non-uniform
 - (D) 4 m/s, at rest
 - (E) 2 m/s, uniform
 - (F) 16 m/s, non-uniform
- 62.** Insert commas correctly in the number 23456789 using the International system. [\[Large Numbers\]](#)
- (A) 234,56,789
 - (B) 2,34,56,789
 - (C) 2,3456,789
 - (D) 23,456,789
 - (E) 2,34,567,89
 - (F) 234,567,89
- 63.** Of 2,000 litres of sludge, 35% is water that gets squeezed out before disposal. How many litres of sludge remain after this water is removed? [\[Wastewater Story\]](#)
- (A) 1,300 L
 - (B) 130 L
 - (C) 1,965 L
 - (D) 1,400 L
 - (E) 1,350 L
 - (F) 650 L
 - (G) 700 L
- 64.** What is the digit in the hundredths place of the number 45.673? [\[A Peek Beyond the Point\]](#)
- (A) 3
 - (B) 0
 - (C) 5
 - (D) 4
 - (E) 7
 - (F) 67
 - (G) 6

65. Car A travels at 72 km/h and car B travels at 22 m/s. Which car is faster, and by how much (in m/s)?

[Motion & Time]

- (A) Car A is faster by 2 m/s
- (B) Car B is faster by 22 m/s
- (C) Car A is faster by 22 m/s
- (D) they are equal
- (E) Car B is faster by 2 m/s
- (F) Car A is faster by 50 m/s
- (G) Car B is faster by 50 m/s

66. How many thousands are there in 45 lakh? [Large Numbers]

- (A) 4,500
- (B) 450,000
- (C) 45,000
- (D) 450
- (E) 4,50,000
- (F) 45
- (G) 4,500,000

67. Why is it dangerous to let untreated sewage flow directly into a river? [Wastewater Story]

- (A) it makes the water safe to drink
- (B) it adds healthy oxygen
- (C) it removes the river's salt
- (D) it cleans the river water
- (E) it spreads diseases and kills aquatic life
- (F) it has no effect at all

68. Which decimal is equal to the fraction $\frac{9}{20}$? [A Peek Beyond the Point]

- (A) 4.5
- (B) 0.20
- (C) 0.9
- (D) 0.55
- (E) 2.22
- (F) 0.45

69. A motorbike travels 144 km in 1 hour 30 minutes. What is its speed in km/h? [Motion & Time]

- (A) 96 km/h
- (B) 72 km/h
- (C) 240 km/h
- (D) 9.6 km/h
- (E) 144 km/h
- (F) 90 km/h
- (G) 110 km/h

70. Write the number "three crore seven lakh nine hundred" in figures. [\[Large Numbers\]](#)

- (A) 3,07,900
- (B) 30,07,00,900
- (C) 3,07,00,090
- (D) 3,70,00,900
- (E) 30,70,900
- (F) 3,07,00,900
- (G) 3,07,09,000

71. Aerators are used in a treatment plant mainly to do what? [\[Wastewater Story\]](#)

- (A) cool the water down
- (B) remove large objects from the sewage
- (C) let sludge settle to the bottom
- (D) pump air into the water so helpful bacteria can break down the waste
- (E) colour the water blue
- (F) add chlorine to the water

72. What is the product 0.04×0.04 ? [\[A Peek Beyond the Point\]](#)

- (A) 0.16
- (B) 0.0008
- (C) 0.0016
- (D) 0.016
- (E) 0.08
- (F) 0.00016

73. A train moves at a steady 50 m/s. How many kilometres does it travel in 2 minutes? [\[Motion & Time\]](#)

- (A) 6.0 m
- (B) 100 km
- (C) 60 km
- (D) 0.6 km
- (E) 6 km
- (F) 3 km
- (G) 6000 km

74. A crore rupees is shared equally among 100 people. How much does each person get? [\[Large Numbers\]](#)

- (A) 10,000
- (B) 1,000
- (C) 1,00,00,000
- (D) 1,00,000
- (E) 50,000
- (F) 10,00,000
- (G) 1,000,00

75. A "vermi-processing" toilet treats human waste using which living organisms? [\[Wastewater Story\]](#)

- (A) fish
- (B) frogs
- (C) algae
- (D) mosquitoes
- (E) bacteria only
- (F) earthworms
- (G) snails

76. What is the value of 6.3 divided by 0.07? [\[A Peek Beyond the Point\]](#)

- (A) 0.44
- (B) 0.011
- (C) 900
- (D) 0.9
- (E) 9
- (F) 90

77. A pendulum is made shorter. What happens to its time period? [\[Motion & Time\]](#)

- (A) the time period doubles
- (B) the time period increases
- (C) the time period decreases
- (D) the period depends only on the bob's weight
- (E) the pendulum stops swinging
- (F) the time period becomes zero
- (G) the time period stays the same

78. Estimate the difference $8,02,156 - 2,97,433$ by rounding each number to the nearest lakh, then subtracting. [\[Large Numbers\]](#)

- (A) 5,00,000
- (B) 6,00,000
- (C) 5,10,000
- (D) 4,00,000
- (E) 5,05,000
- (F) 4,90,000

79. The foul smell of untreated sewage is mainly caused by what? [\[Wastewater Story\]](#)

- (A) bacteria decomposing the organic waste and releasing gases
- (B) the metal of the pipes
- (C) clean oxygen in the water
- (D) chlorine added to the water
- (E) sunlight on the water
- (F) salt dissolved in the water
- (G) the sand and grit in it

80. What number must be added to 6.78 to make 10? [\[A Peek Beyond the Point\]](#)

- (A) 3.78
- (B) 3.22
- (C) 2.22
- (D) 16.78
- (E) 3.32
- (F) 4.22
- (G) 3.12

81. An athlete runs 400 m in 50 seconds, then walks 200 m in 100 seconds. What is the average speed for the whole 600 m? [\[Motion & Time\]](#)

- (A) 2 m/s
- (B) 5 m/s
- (C) 4 m/s
- (D) 600 m/s
- (E) 8 m/s
- (F) 3 m/s

82. A city has 12,00,000 people, and each produces about 150 litres of sewage per day. What is the total sewage produced by the city in one day? [\[Large Numbers\]](#)

- (A) 1,80,00,00,000 L
- (B) 18,00,00,000 L
- (C) 1,80,00,000 L
- (D) 18,00,00,00,000 L
- (E) 12,00,150 L
- (F) 1,80,000 L
- (G) 18,00,000 L

83. A treatment plant processes 9,60,000 litres of sewage steadily in 8 hours. How many litres does it process per minute? [\[Wastewater Story\]](#)

- (A) 1,20,000 L/min
- (B) 200 L/min
- (C) 2,400 L/min
- (D) 20,000 L/min
- (E) 33.3 L/min
- (F) 16,000 L/min
- (G) 2,000 L/min

84. A water tank fills by 0.6 m of height every hour. After how many hours will the water reach a height of 4.5 m? [\[A Peek Beyond the Point\]](#)

- (A) 8 hours
- (B) 0.75 hours
- (C) 2.7 hours
- (D) 7.5 hours
- (E) 7 hours
- (F) 75 hours

85. A car travels at a steady 56.5 km/h. How far does it go in 4.5 hours? [\[Motion & Time\]](#)

- (A) 254.25 km
- (B) 256 km
- (C) 25.425 km
- (D) 254.5 km
- (E) 226 km
- (F) 252 km
- (G) 282.5 km

86. Which is larger: 5 million or 50 lakh? [\[Large Numbers\]](#)

- (A) 5 million is larger
- (B) 50 lakh equals 5 crore
- (C) they are equal (both are 50,00,000)
- (D) 5 million is half of 50 lakh
- (E) 50 lakh is ten times 5 million
- (F) 50 lakh is larger
- (G) 5 million equals 5 lakh

87. Which everyday practice helps reduce the load on a sewage treatment plant? [\[Wastewater Story\]](#)

- (A) putting solid waste in dustbins instead of down the drain
- (B) pouring leftover paint down the sink
- (C) emptying tea leaves into the basin
- (D) pouring cement slurry down the drain
- (E) flushing plastic wrappers
- (F) tipping used cooking oil into the drain
- (G) washing sand into the toilet

88. Evaluate $0.2 \times 0.3 \times 0.5$. [\[A Peek Beyond the Point\]](#)

- (A) 0.1
- (B) 0.6
- (C) 0.05
- (D) 0.3
- (E) 0.030
- (F) 0.003
- (G) 0.03

89. Two stations are 240 km apart. A train leaves at 9:30 am and travels at a steady 80 km/h. At what time does it arrive? [\[Motion & Time\]](#)

- (A) 1:30 pm
- (B) 11:30 am
- (C) 12:30 am
- (D) 12:30 pm
- (E) 12:00 pm
- (F) 1:00 pm
- (G) 3:00 pm

90. Round the number 4,56,789 to the nearest thousand. [\[Large Numbers\]](#)

- (A) 4,56,800
- (B) 5,00,000
- (C) 4,50,000
- (D) 4,57,789
- (E) 4,57,000
- (F) 4,60,000

91. A household uses 0.75 kilolitres (kL) of water per day. How many litres does it use in 4 days? (1 kL = 1000 L) [\[Wastewater Story\]](#)

- (A) 30 L
- (B) 750 L
- (C) 3,000 kL
- (D) 300 L
- (E) 3,000 L
- (F) 3 L
- (G) 2,250 L

92. A car's fuel tank holds 40 litres. It uses 8.5 litres on one trip and 6.75 litres the next day. How much fuel is left? [\[A Peek Beyond the Point\]](#)

- (A) 24.65 L
- (B) 15.25 L
- (C) 25.25 L
- (D) 31.5 L
- (E) 24.85 L
- (F) 24.75 L
- (G) 23.75 L

93. A bullet train covers 270 km in 1.5 hours at a steady rate. Express its speed in metres per second.

[\[Motion & Time\]](#)

- (A) 45 m/s
- (B) 648 m/s
- (C) 25 m/s
- (D) 75 m/s
- (E) 180 m/s
- (F) 500 m/s
- (G) 50 m/s

94. Among 7,07,007, 70,70,070, 7,70,700 and 77,007, which is the smallest? [\[Large Numbers\]](#)

- (A) 7,00,007
- (B) 7,07,007
- (C) 7,70,700
- (D) 70,70,070
- (E) 77,007
- (F) 77,070
- (G) 7,07,070

95. A water bill shows 18.5 kilolitres used, charged at Rs 12.40 per kilolitre. What is the bill amount?

[\[Wastewater Story\]](#)

- (A) Rs 2,294.00
- (B) Rs 229.40
- (C) Rs 22.94
- (D) Rs 230.00
- (E) Rs 30.90
- (F) Rs 228.00
- (G) Rs 229.04

96. Which is larger, $\frac{3}{5}$ or 0.55, and by how much? [\[A Peek Beyond the Point\]](#)

- (A) 0.55 is larger by 0.15
- (B) 0.55 is larger by 0.05
- (C) $\frac{3}{5}$ is larger by 0.15
- (D) $\frac{3}{5}$ is larger by 0.5
- (E) they are equal
- (F) $\frac{3}{5}$ is larger by 0.05
- (G) 0.55 is larger by 0.5

97. On a distance-time graph, line P is steeper than line Q. P reaches 100 m in 10 s and Q reaches 100 m in 25 s. Which is faster, and what is P's speed? [\[Motion & Time\]](#)

- (A) P is faster, at 25 m/s
- (B) they move at the same speed
- (C) Q is faster, at 10 m/s
- (D) P is faster, at 10 m/s
- (E) Q is faster, at 4 m/s
- (F) P is faster, at 100 m/s

98. The population of a town is 8,40,000. If 1,67,500 of them are children, how many are adults? [\[Large Numbers\]](#)

[Numbers\]](#)

- (A) 6,72,500
- (B) 7,72,500
- (C) 10,07,500
- (D) 6,82,500
- (E) 6,72,000
- (F) 6,73,500
- (G) 5,72,500

99. In a wastewater treatment process, which statement is TRUE about helpful bacteria? [\[Wastewater Story\]](#)

- (A) they need oxygen (from aerators) to break down the dissolved waste
- (B) they settle out as clean drinking water
- (C) they work best with no air at all in the aeration tank
- (D) chlorine is added to help them grow
- (E) they are removed by the bar screens
- (F) they make the water dirtier
- (G) they are the same as the grit and sand

100. A shopkeeper buys 2.5 kg of apples at Rs 80 per kg and 1.5 kg of grapes at Rs 60 per kg. What is the total bill? [\[A Peek Beyond the Point\]](#)

- (A) Rs 350
- (B) Rs 300
- (C) Rs 90
- (D) Rs 290
- (E) Rs 280
- (F) Rs 230
- (G) Rs 200